1 **(a)** **(i)** 47.54 **[B1]**

(ii) 48 **[B1]**

(iii) 48 **[B1]**

[Total Marks: 3]

2 **(a)** 780 **[B1]**

(b) 0.036 **[B1]**

(c) 4.10 **[B1]**

[Total Marks: 3]

3 **(a)** 48.98 **[B1]**

(b) 0.0031 **[B1]**

(c) 600 **[A1]**

M1 for $\frac{8 \times 30}{0.4}$

[Total Marks: 4]

4 **(a)** **(i)** 34.57 **[B1]**

(ii) 0.070 **[B1]**

(b) $\frac{60 \times 4}{0.5}$ **[M1]**

480 **[B1]**

[Total Marks: 4]

5 **(a)** 0.048 **[B1]**

(b) 380 **[B1]**

(c) $\frac{60 \times 4}{20 - 3}$ or $\frac{240}{17}$ **[M1]**

14 or 14.1 (from $\frac{240}{17}$) or 15 (if 17 is estimated as 20) **[A1]**

[Total Marks: 4]

6 **(a)** **(i)** 203.50 **[B1]**

(ii) 203 **[B1]**

(iii) 200 **[B1]**

(b) 560 **[A1]**

If 0 scored, M1 for $\frac{7 \times 40}{0.5}$ **[M1]**

[Total Marks: 5]

7 $\frac{60 \times 3}{0.2}$ **[M1]**

$\frac{180}{0.2}$ **[M1]**

900 **[A1]**

[Total Marks: 3]

8 (a) $\frac{8 \times 30}{0.4}$ **[M1]**

600 **[A1]**

(b) $\sqrt{100}$ or 10, and 20, and 0.5 **[M1]**

$\frac{10 + 20}{0.5^2}$ or $\frac{30}{0.25}$ **[M1]**

120 **[A1]**

[Total Marks: 5]

9 (a) $\frac{\sqrt{100} \times 4^2}{0.2}$ **[M1]**

$\frac{10 \times 16}{0.2}$ or $\frac{160}{0.2}$ **[M1]**

800 **[B1]**

(b) 500×200 **[M1]**

100000 **[B1]**

[Total Marks: 5]

10 (a) 2000 **[A3]**

If 0 scored, M1 for rounding the numbers to 4×10^6, 0.05 and 20 **[M1]**

M1 for evaluating the numerator to 2000 and the denominator to 1 **[M1]**

(b) Greater (than) **[B1]**

Valid reason, e.g. the numerator was rounded up and the denominator was rounded down. **[B1]**

[Total Marks: 5]

11 (a) 1600 **[A1]**

M1 for $\frac{\sqrt[3]{8} \times 20^2}{0.5}$

M1 for 2×400 or 800 seen in numerator

(b) 3.2×10^6 **[A1]**

M1 for $4 \times 10^5 \times 2^3$

M1 for 32×10^5 or 3200000 seen

[Total Marks: 6]

12 (a) $\sqrt{100} \times 4^3$ or 10×64 $(= 640)$ **[M1]**

0.2×50 $(= 10)$ **[M1]**

$\dfrac{10 \times 64}{10} = 64$ **[A1]**

(b) Uses $a = 4$, $x = 0.05$, $b = 1$, $c = 0.8$ **[M1]**

$\dfrac{4 \times 0.05}{1 - 0.8}$ or $\dfrac{0.2}{0.2}$ **[M1]**

1 **[A1]**

[Total Marks: 6]

13 (a) 40, 5, and 1000 seen **[M1]**

$\dfrac{40 \times 5^2}{\sqrt[3]{1000}}$ or $\dfrac{40 \times 25}{10}$ **[M1]**

100 **[A1]**

(b) Radius ≈ 4 and Height ≈ 10 **[B1]**

Volume $= 3 \times 4^2 \times 10$ **[M1]**

480 **[A1]**

(c) Distance ≈ 4000 and Speed ≈ 600 **[B1]**

$\dfrac{4000}{600}$ or $\dfrac{20}{3}$ or 6.66... hours **[M1]**

6 hours and 40 minutes **[A1]**

[Total Marks: 9]

14 (a) $125 \times 74 = 9250$ **[M1]**

9300 **[A1]**

(b) $x^2 = 9250$ **[M1]**

$x = \sqrt{9250}$ (96.176...) **[M1]**

96.2 **[A1]**

[Total Marks: 5]

15 (a) 404 **[A2]**

If 0 scored, M1 for $\frac{1}{3} \times \pi \times 5.82^2 \times 11.4$ **[M1]**

(b) 360 **[A2]**

If 0 scored, M1 for $\frac{1}{3} \times 3 \times 6^2 \times 10$ **[M1]**

(c) 70 **[A2]**

If 0 scored, M1 for $850 \div 12$ or 70.83... seen **[M1]**

[Total Marks: 6]

16 (a) 87.1 **[B2]**

M1 for 87.098... seen or $\frac{78.38...}{0.9}$

(b) (i) 27 **[A1]**

M1 for $\frac{4}{3} \times \pi \times 1.85^3$ or 26.5... seen

(ii) 35 hours 22 minutes **[A1]**

M1 for their exact volume (or 27) $\times 1000$ (e.g. 26522 litres)

M1 for $\frac{\text{their } 26522}{12.5}$ $(= 2121.76$ mins$)$

[Total Marks: 7]

17 (a) (i) 70.35465... **[B1]**

Accept an answer containing at least 5 decimal places.

(ii) 70.4 **[B1]**

(b) $\frac{30^2 - 10^2}{\sqrt{100}}$ or $\frac{900 - 100}{10}$ **[M1]**

80 **[B1]**

[Total Marks: 4]

18 (a) $r \approx 4$, $h \approx 20$, density ≈ 8 **[M1]**

Estimated Mass $= 3 \times 4^2 \times 20 \times 8$ **[M1]**

7680 **[A1]**

(b) Volume $= \pi \times (4.15)^2 \times 19.8 = 1071.3\ldots$ **[M1]**

Mass $= 1071.3 \cdots \times 8.13 = 8710.19\ldots$ **[M1]**

8710 **[A1]**

[Total Marks: 6]